

Umang Desai

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Summary

Senior Software Engineer with over 12 years of experience designing, developing, and deploying scalable, AI-driven systems. Currently building agentic AI services powered by LLMs, Claude SDK, and Model Context Protocol (MCP) — with expertise in autonomous planning, tool orchestration, and human-in-the-loop workflows. Deep background in distributed systems, cloud-native architectures, and production observability using Python, Golang, and OpenTelemetry. Hands-on, versatile engineer with an eye for simplicity and elegance.

Skills

Languages: Python, Golang, TypeScript, Java, C/C++, SQL

AI & LLM: Claude SDK, Model Context Protocol (MCP), Agentic AI Systems, LLM Integration (Azure OpenAI, Bedrock), Prompt Engineering, NLtoSQL, Natural Language Understanding (NLU)

Infrastructure & Messaging: Docker, Kubernetes, Redis (Pub/Sub, Caching), gRPC, SSE (Server-Sent Events), RabbitMQ, Kafka

Cloud Platforms: AWS, GCP, Azure

Observability: OpenTelemetry (Traces, Spans, Logs SDK), Prometheus, Grafana, Fluentd, ElasticSearch

Databases: CockroachDB, AuroraDB, MySQL, MongoDB, SQLite, Redis

DevOps & Security: Git, Jenkins, CI/CD, CodeQL, Docker Security Hardening, Sandboxed Code Execution

Practices: AI-Driven Development, Distributed Systems Design, Agile/Scrum, Architectural Decision Records (ADRs), Performance Optimization

Work Experience

Cisco

July 2023 – Present

Senior Software Engineer

- Lead design and development of Canvas Deep Agent, Cisco's agentic AI service on the AI Studio platform — building an autonomous LLM-powered agent using Python, Claude SDK, and Model Context Protocol (MCP) that plans, reasons, and executes multi-step tasks with human-in-the-loop approval gates.
- Pioneer AI-driven development practices across the engineering team, leveraging LLM-powered tooling to accelerate feature development, debugging, and architectural decision-making — maintaining structured audit trails with engineering decision records, session-tracked investigations, and traceable design-to-deployment workflows.
- Architect distributed systems infrastructure for multi-pod Kubernetes deployments, implementing Redis pub/sub for cross-pod communication, plan approval gating mechanisms, and resilient session management that gracefully degrades under failure.
- Build end-to-end OpenTelemetry observability for the agent lifecycle — instrumenting planning loops, tool execution chains, and LLM token utilization with granular spans and durable log export, providing full audit trails of every agent action for production debugging, performance optimization, and compliance.
- Drive security hardening initiatives including Docker image hardening, CodeQL remediation, sandboxed code execution with network isolation, and architectural SQL injection mitigation.
- Develop Cisco's AI Assistant for FMC and CDFMC with NLtoSQL capabilities, translating natural language queries into precise SQL for firewall management, achieving a 70% latency reduction through query optimization, caching, and indexing.
- Refactor ML prototypes into production-ready backend microservices using Python, Golang, CockroachDB, and AuroraDB, with feature flags, resiliency patterns, and comprehensive error management for enterprise-grade deployment.

Armorblox

May 2021 – July 2023

Senior Software Engineer

- Designing and developing Armorblox Data Protection using Regex and NLU (Natural Language Understanding) in GoLang, C and / or Python.
- Applying distributed system concepts like RabbitMQ, Kafka and Kubernetes in scaling the Armorblox Data Protection product to process large volumes of data in GCP and Azure cloud data centers.
- Designing and implementing configuration and incident management using SQL data stores and gRPC.
- Collaborating with cross functional teams like Product Management, Customer Support and Production Engineering for successful delivery of the product and best customer experience.
- Improve the code quality by resolving software defects and adding unit and integration test cases.

Kazuhm, Inc.

Aug 2018 – April 2021

Senior Software Engineer

- *Maintained and re-engineered* current Windows development stack (*C++*, *Visual Studio*, *WIX*).
- *Lead Migrating* Windows stack to newer technologies to enable cross-platform compatibility and optimized architecture across Kazuhm platform. (*GoLang*)
- *Designed, developed and tested a distributed and scalable* container deployment system (*Docker*) for distributed workloads using the Kazuhm platform.
- *Architected and configured platform-wide* logging framework (*Fluentd*) along with *ElasticSearch/S3* as storage and Grafana as UI.

George Mason Univ. Department of Computer Science

Jan 2017 – May 2018

Graduate Teaching Assistant (Software Engineering)

- Conducted lectures, labs, and mentored students on SDLC practices, Agile culture, and Java OOP fundamentals. Graded projects, assignments, and peer evaluations. Introduced a culture of unit testing.

Fondo

Jan 2015 – Mar 2016

Co-Founder

- Co-founded social event discovery platform; Designed backend, databases, and data models.

ESQ Business Services

Aug 2012 - Dec 2014

Software Engineer

- Designed, developed and tested module to *monitor JVM* on a host by *extracting telemetry data* using *JAVA*, *MSSQL* and *JMX*. This was achieved via a 3 tier *client-server-data backend architecture*.
- Designed and modeled a database for the data extracted. Then *developed advanced filtering* across *multiple Mbeans* for retrieving these metrics for *bandwidth friendly data pipelines*.
- Configured frameworks like *XStream* for configuration files and models, *log4j* for logging.
- Designed, developed and tested module to *monitor Linux via the proc fs to analyze system performance and uptime*. This was achieved with *JAVA* and *MSSQL*
- Developed a *parsing framework for system proc files with configurable polling mechanisms*.

Education

M.S Computer Science

2016-2018

George Mason University

GPA: 3.4

B.S Information Science and Technology

2008-2012

Pennsylvania State University

GPA: 3.0

Volunteer Experience

Pacific Youth Soccer League(PYSL Blast)

Soccer Head Coach

2026 - Present

SDSC Surf

Soccer Head Coach

2020 -2024

Personal Projects

Content Addressable Network.

- Built a *scalable distributed hash table network*. The idea is to help manage a large hash table over the Internet like scalability. This can also support distributed data storage. Used: Java, RMI

Soccer Fitness Tracker (<https://tinyurl.com/bdhmyjrr>)

- iOS application for tracking and analyzing soccer-specific fitness metrics, training loads, and player performance over time. Free and Community driven.

Tryout Assistant (<https://tinyurl.com/827m6752>)

- iOS application to help coaches take hands free notes during tryouts. Organize, evaluate, and manage player tryouts with structured scoring, notes, and roster-building tools.

Touchline Studio (<https://tinyurl.com/y9jdc9td>)

- iOS application for soccer coaches to plan training sessions, design drills, and manage team preparation. Development tree's, Lineup management. Free and Community driven.

Operating System

- Operating systems fundamental development. Implemented OS concepts like synchronization primitives, process creation, multi-processing, virtual memory. Used: C, gcc, vi